

Name: _____



Making a journey

Task A: Calculating journey times for different transport

1) The table shows the distances of three **commutes**.

Fill in the blanks in the table by calculating the estimated journey times using each method of transport.

method of transport	average speed (km/h)	estimated journey times (to the nearest minute)		
		2 km journey	8 km journey	16 km journey
walk	5			
cycle	20			
drive	35			

Which method of transport would you choose to use for each length of journey?

Task B: Calculating carbon emissions for journeys

The amount of CO₂ emissions per mile can be estimated by using the formula:

$$\text{emissions per mile (grams)} = \frac{\text{CO}_2 \text{ per gallon of fuel (grams)}}{\text{fuel efficiency of the vehicle (mpg)}}$$

1) A person lives 6 miles from their place of work and travels by car.

The average CO₂ emissions for a petrol car is 9 kg per gallon.

On average, their car travels 45 miles per gallon.

They work 223 days per year.

Calculate the amount of CO₂ emissions produced per year from **commuting** by car.

Give your answer in kilograms and accurate to 3 significant figures.

2) 40 people live 6 miles from their place of work and travel on the same bus.

The average CO₂ emissions for a diesel bus is 10 kg per gallon.

On average, the bus travels 8 miles per gallon.

They work 223 days per year.

Calculate the amount of CO₂ emissions produced per passenger per year from

commuting by bus. Give your answer in kilograms and accurate to 3 significant figures.



Task C: Calculating the cost of a journey

- 1) A person lives 6 miles from their place of work and travels by car 225 days per year.
On average, their car travels 45 miles per gallon.
At their nearest petrol station, petrol is sold for 148.9 p per litre.
1 gallon $\approx$ 4.546 litres.
- a) How much do they spend on petrol per year for **commuting** by car?
- b) Their car insurance costs £65 per month.
Their annual maintenance costs sum to £820.
Assume that they only use the car for **commuting**.
How much do they spend in total on their car each year?
- c) They are considering selling their car and **commuting** by bus.
A weekly bus pass in their town costs £22 per week.
Assume that they would buy a weekly bus pass every week of the year.
Would **commuting** by bus cost more or less money per year? Justify your answer.